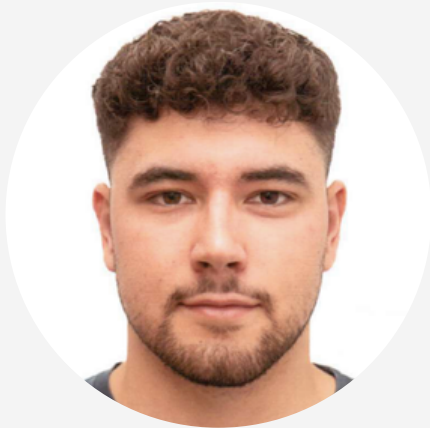




Javier Gutiérrez

Test Method Development & Validation Engineer

CONTACT

- +34 601 27 50 55
- jgutierrezr.2003@gmail.com
- [linkedin.com/in/jgutierrezr2003](https://www.linkedin.com/in/jgutierrezr2003)
- Valladolid (Spain)

SKILLS

- Advanced knowledge of:
 - AVL Puma Open 2
 - AVL Navigator
 - Solid Edge
 - Abaqus
 - MATLAB
 - ANSYS
 - Python
- Mechanical design and analysis
- FEM structural analysis
- Knowledge of powertrain and propulsion systems

COMPETENCIES

- Strong teamwork skills
- Analytical thinking
- Problem-solving abilities
- Continuous improvement mindset

LANGUAGES

- Spanish | Native
- English | Upper-intermediate
 - B2 First (CFE)

OTHERS

- Driving licence (A2 - B)
- Own vehicle available
- Flexible schedule

PROFILE

I'm a Mechanical Engineer graduated from Universidad Carlos III de Madrid, specialized in propulsion systems. During my studies, I developed a strong foundation in mechanical design, structural analysis, manufacturing processes, and numerical simulation, gaining proficiency in CAD tools, Computational Fluid Dynamics (CFD), Finite Element Analysis (FEA), and mechanical system optimization.

Currently, I work as an engineer in the Methodology Department at HORSE R&D, through CT Engineering Group, within the engine test bench area. My role focuses on the development, validation, and optimization of testing methodologies for their implementation in internal combustion engines (ICE) and hybrid powertrains.

I consider myself a determined, adaptable, and committed professional, driven by continuous learning, eager to take on new challenges, and motivated to contribute value from day one.

EXPERIENCE

ICE and Hybrid Engine Test Facilities & Methods Specialist — HORSE R&D&I

Nov 2025 - Present

- Development and implementation of durability test methods for mechanical validation of ICE, mild-hybrid and hybrid engines.
- Development and implementation of calibration and tuning methods for ICE, mild-hybrid and hybrid engines.
- Technical definition, configuration and commissioning of engine and powertrain test beds in both new development and upgrade projects.

Image processing & analysis — UC3M

Jan 2025 - Sept 2025

- Development of MATLAB code to apply different image processing techniques to videos obtained from hydrogen and methane combustion experiments carried out in a Hele-Shaw cell.
- Implementation of advanced digital processing functions focused on edge detection.

Structural analysis and design using Abaqus — UC3M

Sept 2025 - Dec 2025

- Design of a single-seater chassis in Abaqus, meeting the required dimensional and weight constraints.
- Finite Element Analysis (FEM) was performed to evaluate torsional strength, and the design was iteratively optimized based on the results.

EDUCATION

Bachelor's Degree in Mechanical Engineering

Sept 2021 - Sept 2025

Universidad Carlos III de Madrid | Leganés

High School Diploma in Science and Technology

Sept 2019 - Jun 2021

IES Padre Juan de Mariana | Talavera de la Reina